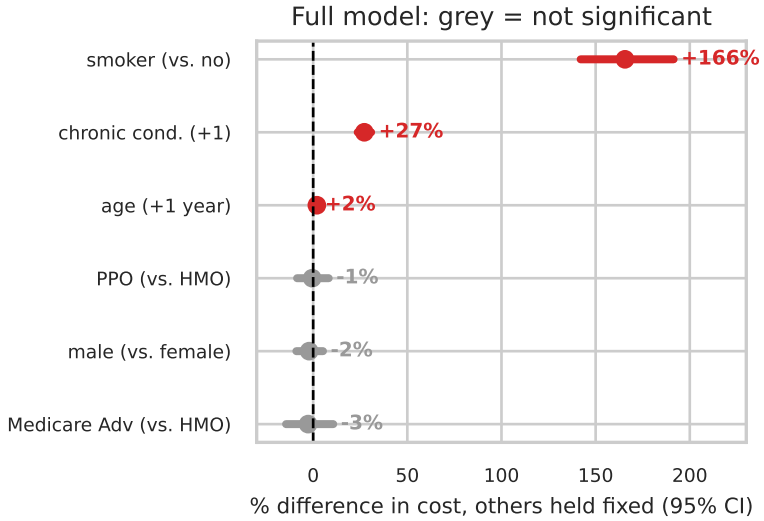
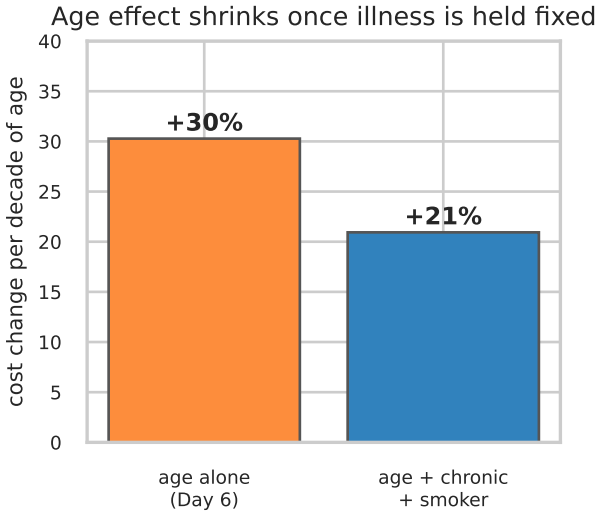
Day 7 Cheat Sheet: Multiple Regression & Dummy Variables

Several predictors at once: each coefficient is 'holding the others fixed'

TERM	WHAT IT MEANS	CODE
Multiple regression	one line, many x's: y = b0 + b1 x1 + b2 x2 + ...	smf.ols("y ~ x1 + x2", df)
"Holding fixed"	b1 = change in y for +1 x1 when x2.. don't move	model.params
Dummy variable	0/1 column per category level, minus one	C(plan, Treatment('HMO'))
Reference level	the level every dummy is compared against	pd.get_dummies(drop_first=True)
Adjusted R squared	R squared with a penalty per extra column	model.rsquared_adj
Multicollinearity	predictors that move together -> shaky coefs	variance_inflation_factor
Held-out R squared	how well it predicts patients it never saw	train_test_split + r2_score

LOG COST AGAIN: exp(coef) - 1 is the % difference. Smoker coef 0.977 -> exp(0.977) - 1 = +166%





THIS DATA (simulated, reused from Days 5-6): 2,000 patients, outcome = log(annual cost)
Age slope: 0.0264 alone, 0.0190 after adding chronic + smoker (+30% vs. +21% per decade)
Full model: smoker +166%, each chronic condition +27%, each year +2.0%; plan and sex not significant (p > 0.5)
R squared 0.233 (age only) -> 0.436 (age + chronic + smoker); adjusted R squared 0.4349 -> 0.4342 after adding plan + sex
Adding BMI silently drops 142 rows with missing BMI (n = 1858) | test R squared 0.22 (age) vs. 0.45 (6 features)

HABITS THAT KEEP YOU HONEST
- Say "holding the others fixed" every time you read a multiple-regression coefficient
- Pick the reference level on purpose (the usual or largest group) and name it in your write-up
- Compare adjusted R squared, not R squared, when models have different numbers of columns
- Check n after every fit: missing values quietly shrink your sample
- Not significant is not 'no effect': read the CI (plan: about -14% to +11%)

```
m = smf.ols("log_cost ~ age + chronic_conditions + C(smoker) + C(plan, Treatment('HMO'))", df).fit()
print(m.summary()); print(np.expm1(m.params)); print(m.rsquared_adj, m.nobs)
X = pd.get_dummies(df[["age","smoker","plan"]], drop_first=True, dtype=int)   # for scikit-learn
X_tr, X_te, y_tr, y_te = train_test_split(X, df["log_cost"], test_size=.25, random_state=42)
```

Infographic: One Predictor to Many

Simulated: 2,000 patients, log(annual cost) on age, illness, smoking, plan and sex

Start	Add	Encode	Read	Penalize	Test
Day 6 line: cost ~ age	chronic + smoker	categories to 0/1 dummies	% change, others fixed	adjusted R squared	unseen patients

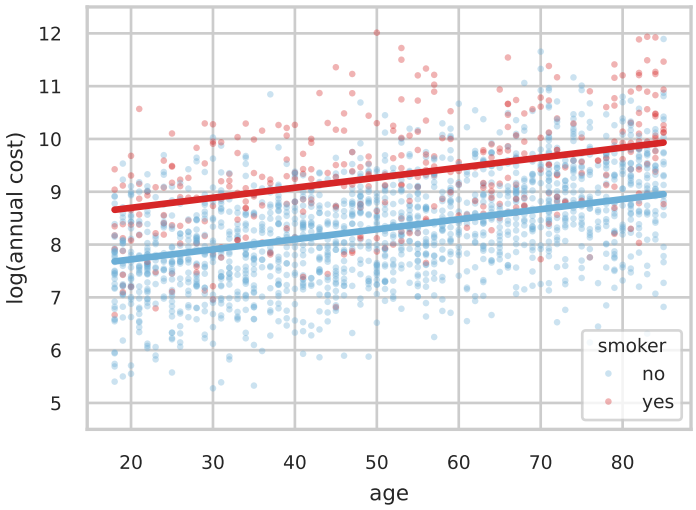
R SQUARED
0.23 to 0.44

SMOKER
+166% cost

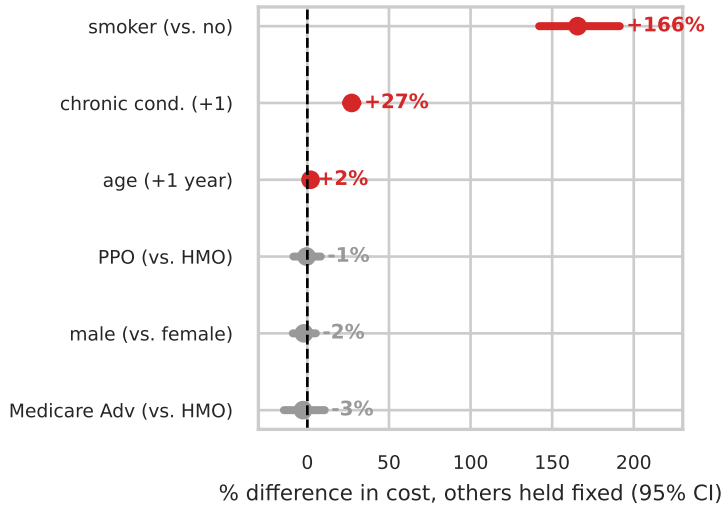
PER CONDITION
+27% cost

TEST R SQ
0.22 to 0.45

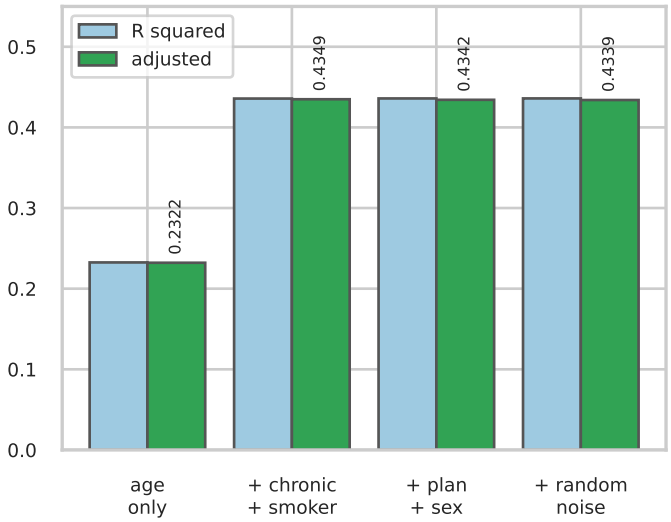
Dummy = parallel lines, shifted up for smokers



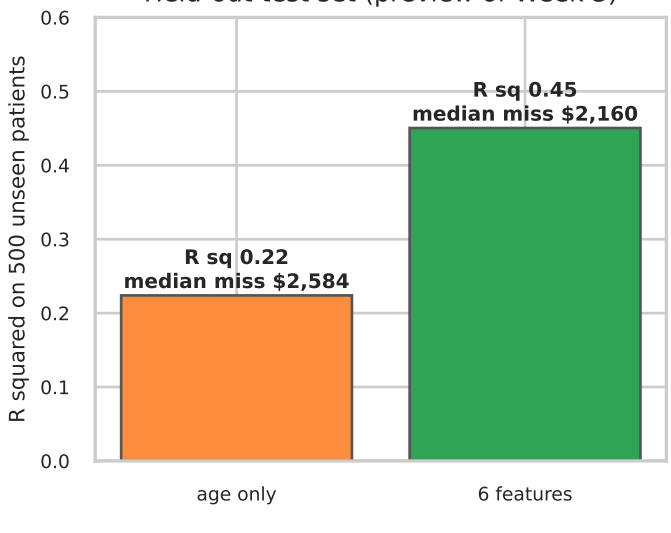
Full model: grey = not significant



Adding columns: R sq creeps up, adjusted does not



Held-out test set (preview of Week 3)



Why the age slope shrank: older patients have more chronic conditions (r = 0.40). Alone, age gets credit for that illness. Once chronic conditions are in the model, age's own share is +21% per decade instead of +30%.

Hands-On Worksheet: ds-genai-90-days / day07

60-90 minutes | notebook: day07/multiple_regression_patients.ipynb

[ ] mkdir day07, copy day06/patients.csv in, conda activate ds90

[ ] Load, drop duplicates (2,000 rows), add log_cost; count missing BMI

[ ] Fit log_cost ~ age, then ~ age + chronic_conditions + smoker

[ ] Compare the age slope and R squared. Why did the slope shrink?

[ ] Full model with C(smoker), C(plan, Treatment('HMO')), C(sex)

[ ] Build the % change table: np.expm1(params) and np.expm1(conf_int())

[ ] Which coefficients are significant? Read the CI for plan, not just p

[ ] Add plan + sex, then a random noise column: watch R sq vs. adjusted

[ ] Add bmi: check model.nobs. How many patients disappeared?

[ ] VIF for every predictor: any above 5?

[ ] Train/test split: age-only vs. 6 features, test R sq and median $ miss

[ ] Findings cell; git add day07 && git commit && git push

[ ] Stretch: Snowflake REGR_SLOPE by smoker group + IFF() dummies

```
import numpy as np, pandas as pd, statsmodels.formula.api as smf
df = pd.read_csv("patients.csv").drop_duplicates()
df["log_cost"] = np.log(df["annual_cost"])
m = smf.ols("log_cost ~ age + chronic_conditions + C(smoker) "
            "+ C(plan, Treatment('HMO')) + C(sex)", data=df).fit()
print((np.expm1(m.params)).round(3))           # expect smoker ~ +1.66, chronic ~ +0.27
print(m.rsquared, m.rsquared_adj, m.nobs)        # expect ~0.436, ~0.434, 2000
print(smf.ols("log_cost ~ age + bmi", df).fit().nobs)   # fewer rows: missing BMI
```

Age slope alone: _____ multiple: _____ Smoker %: _____ Chronic %: _____

Adj R sq (3 cols): _____ (full): _____ n with BMI: _____ Test R sq: ____ vs ____

Day 7 Self-Check Quiz

1. The age slope drops from 0.0264 to 0.0190 after adding chronic conditions. Why?

_____

2. The smoker coefficient on log cost is 0.977. Translate it for a manager.

_____

3. Plan has three levels (HMO, PPO, Medicare Adv). How many dummy columns, and why?

_____

4. Plan's p-value is 0.68. Can you say 'plan has no effect on cost'?

_____

5. R squared rose when you added a random noise column. What should you compare instead?

_____

6. Adding BMI changed n from 2,000 to 1,858. Why does that matter for comparing models?

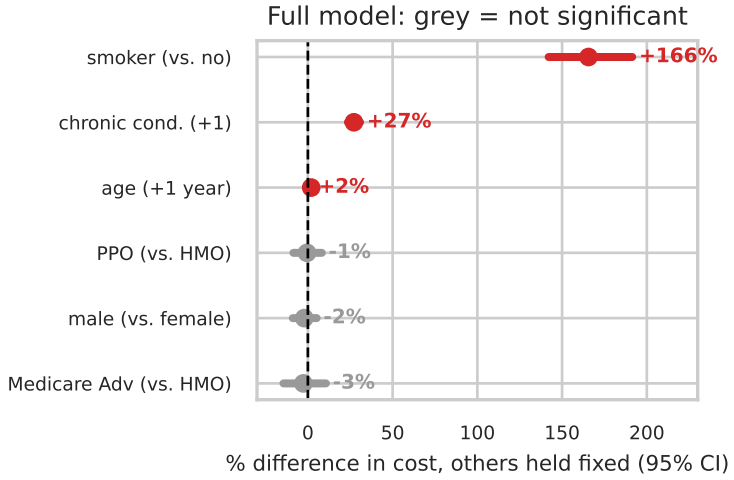
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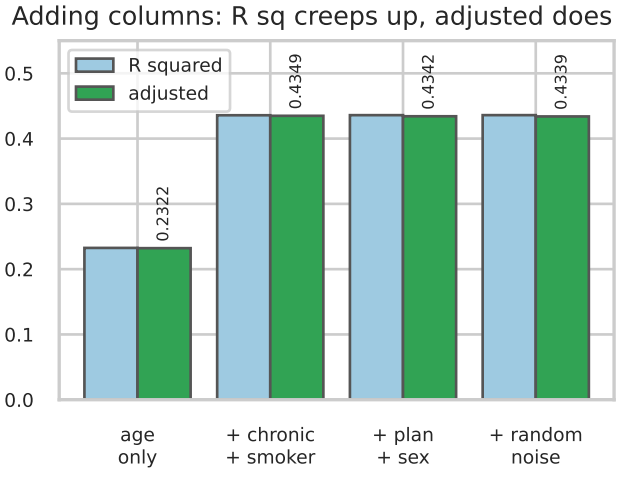
7. What is a VIF, and what value would worry you?

_____

8. Why judge the model on a held-out test set instead of the training R squared?

_____





Answers:
1 older patients have more chronic conditions (r = 0.40); alone, age took credit for illness. Now it is age 'holding chronic and smoking fixed'
2 "Smokers cost about 2.7 times as much (+166%, 95% CI +142% to +191%) as non-smokers of the same age and illness burden"
3 two: HMO is the reference; a third column would be perfectly predictable from the others (the dummy trap)
4 no: it is not significant, but the CI (about -14% to +11%) still allows small real differences. Say 'no clear difference'
5 adjusted R squared (it fell from 0.4342 to 0.4339), or held-out performance 6 the models use different patients; compare on the same rows
7 variance inflation factor: how much a coefficient's variance grows from overlap with other predictors; above 5 look, above 10 worry
8 training R squared rewards memorizing; test R squared (0.45 here) shows how it does on new patients